

Basic Principles of Medicine 2
Module: Digestion and Metabolism
Lecture 26

Exocrine Pancreas and Intestinal Transports

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Recommended Reading

Rhoades, R. & Bell, D. (2023); *Medical Physiology: Principles for Clinical Medicine*, 6th Edition; Wolters Kluwer. Chapter 25

[Gastrointestinal Secretory Functions | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library](#)

Objectives

SOM.MK.II.BPM2.3.DM.1.PHYS.1128	List the major components secreted by the exocrine pancreas and the principal cell types involved in this secretion.
SOM.MK.II.BPM2.3.DM.1.PHYS.1129	Describe the mechanisms by which chyme from the stomach is neutralized in the duodenum.
SOM.MK.II.BPM2.3.DM.1.PHYS.1130	List the stimuli that release secretin and CCK and explain the route by which these regulatory peptides stimulate the pancreas.
SOM.MK.II.BPM2.3.DM.1.PHYS.1131	Describe the role of CFTR in pancreatic ductal secretion, and predict the consequences of cystic fibrosis on the GI system.
SOM.MK.II.BPM2.3.DM.1.PHYS.1132	Describe the mechanisms by which HCO_3^- is taken up by pancreatic ductal cells
SOM.MK.II.BPM2.3.DM.1.PHYS.1133	Describe the mechanisms whereby the gall bladder concentrates bile, and the endocrine mechanism stimulating gall bladder contraction and the secretion of bile through the sphincter of Oddi into the small intestine.
SOM.MK.II.BPM2.3.DM.1.PHYS.1134	Describe the location and the mechanisms that mediate the intestinal trans-epithelial movement of water, the major electrolytes, iron and calcium.

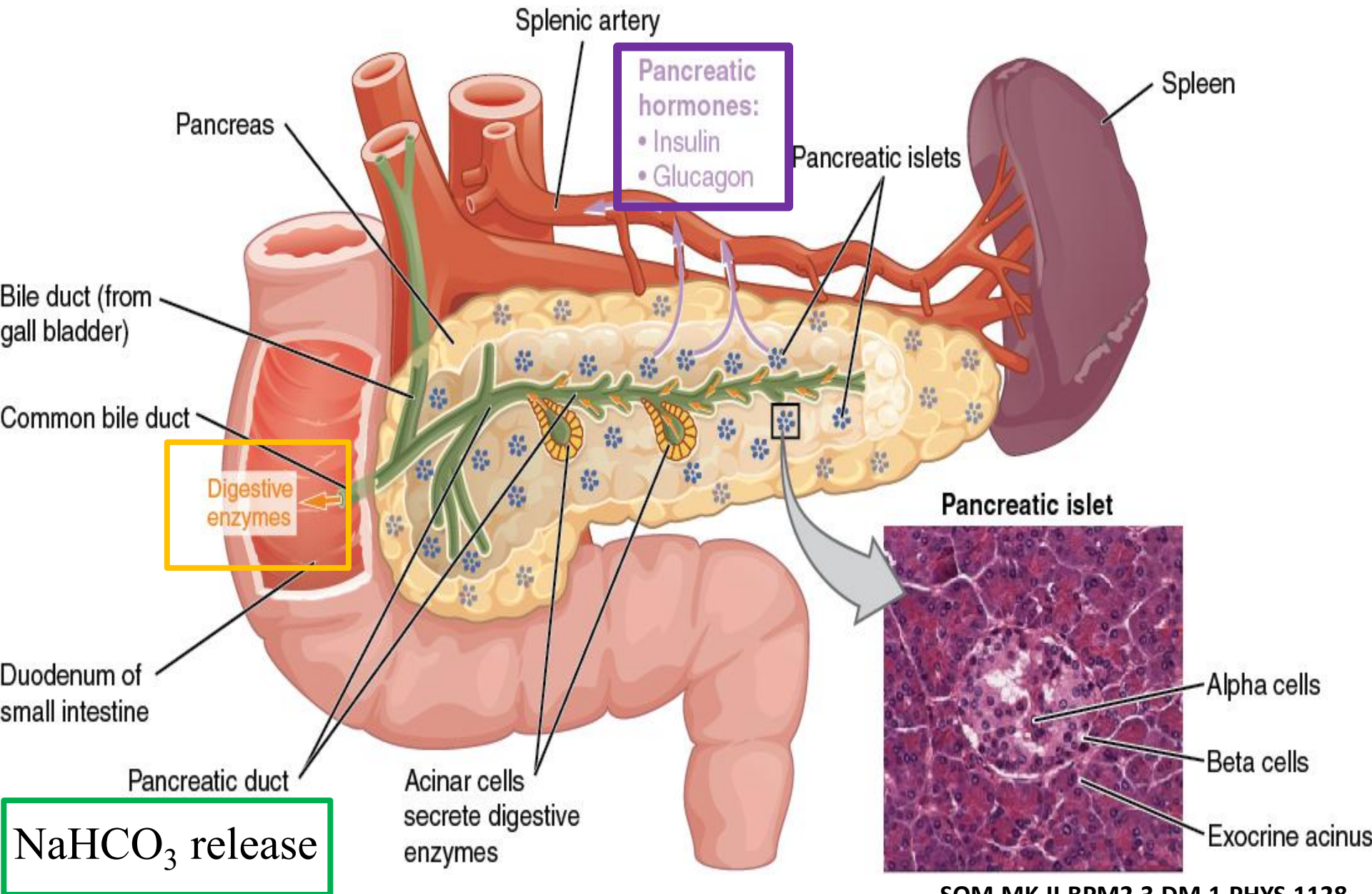
Outline

- Pancreatic exocrine secretions
- Intestinal secretions & absorption
- GI fluid balance and diarrhea
- Summary

Part 1

Pancreatic exocrine secretions

Pancreatic secretions ~1L/day



Pancreatic Secretions

- Functions
 - Digestion
 - Protection
 - Neutralizes the acidity of the contents leaving the stomach
 - Optimal pH for enzyme activity
- Composition:
 - Same concentration of Na^+ and K^+ as plasma - isotonic
 - high HCO_3^- concentrations (alkaline)
 - low Cl^- concentration
 - Enzymes: lipase, amylase and proteases
 - Composition depends on flow rate:
 - Low flow rates: isotonic – mostly Na^+ and Cl^-
 - High flow rates: isotonic – mostly Na^+ and HCO_3^-

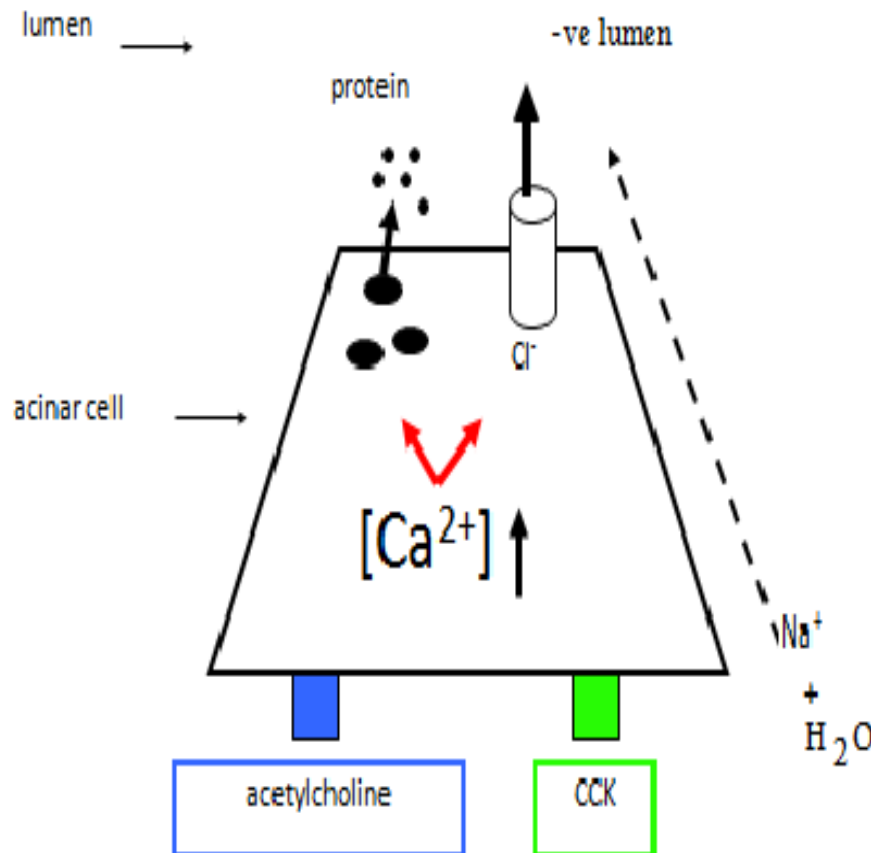
Pancreatic Secretions

- Acinar cells: low volume, NaCl, enzymes
- Ductal cells: modifies; absorbing Cl^- , secreting HCO_3^- via Cl/HCO_3^- exchanger
- Pancreatic ductal cells are permeable to water

Stimuli	Secreted from	In response to	Effect
Secretin	S cells of duodenum	H^+ in duodenal lumen	$\uparrow \text{HCO}_3^-$ secretion
CCK	I cells of duodenum	Small peptides, amino acids, fatty acids in duodenal lumen	$\uparrow$ Enzyme secretion $\uparrow$ Potentiates HCO_3^- secretion
ACh	Vagus terminals	H^+ , small peptides, amino acids, fatty acids	$\uparrow$ Enzyme secretion $\uparrow$ Potentiates HCO_3^- secretion

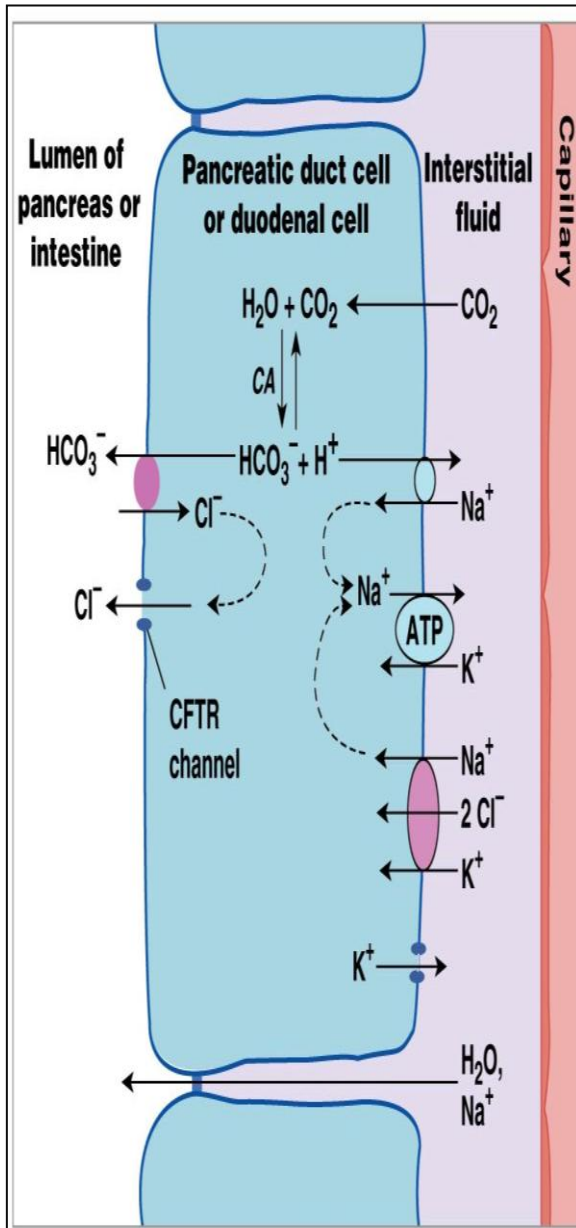
Modulators of pancreatic acinar secretions

A Rise In $[Ca^{2+}]$ Triggers Acinar Cell Secretion



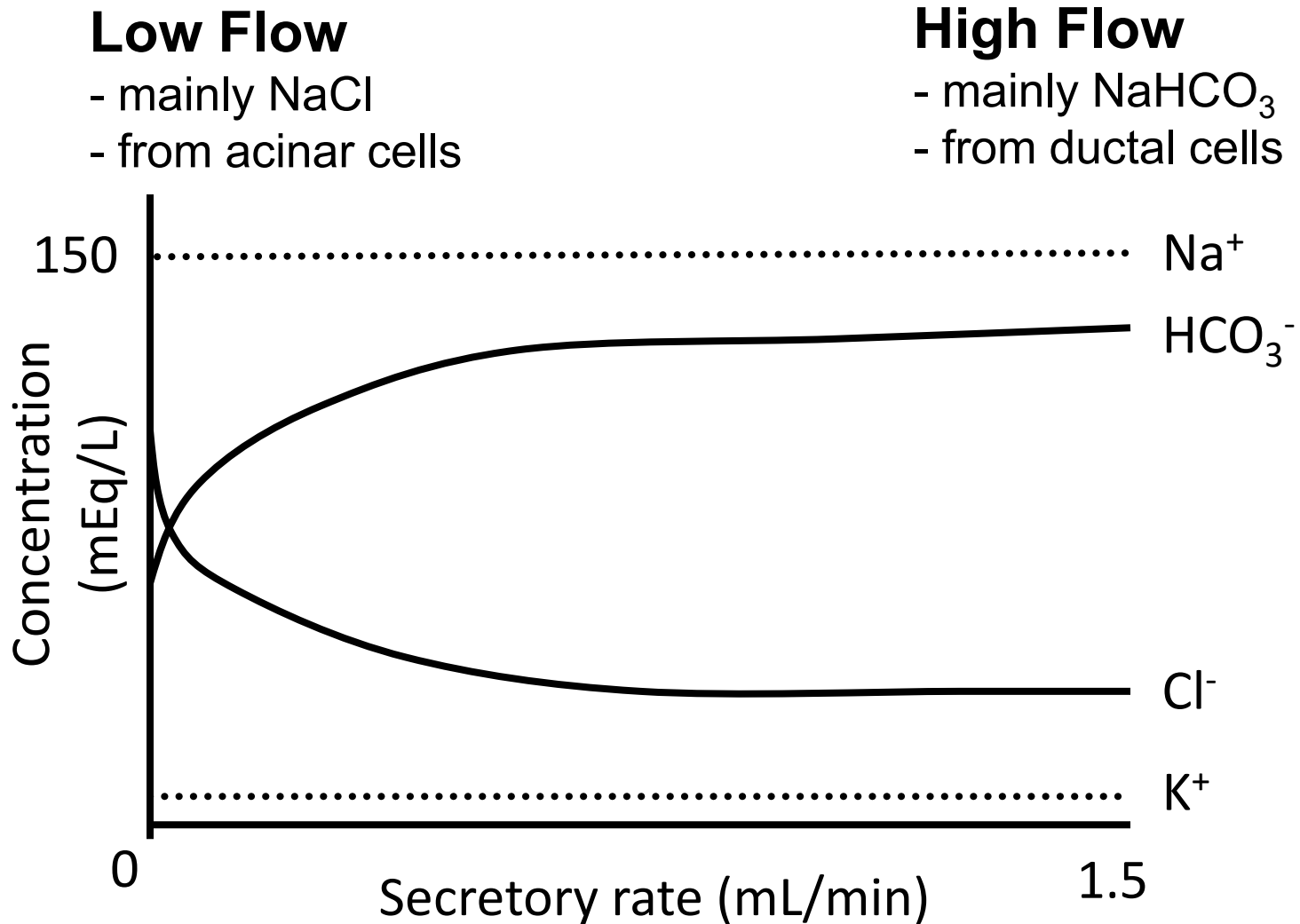
- $\uparrow$ Cl^- secretion = -ve lumen
- $\uparrow$ Na and H_2O follow passively into the lumen
- $\uparrow$ Exocytosis - release of zymogen granules into SI lumen

Bicarbonate secretion by Ductal cells

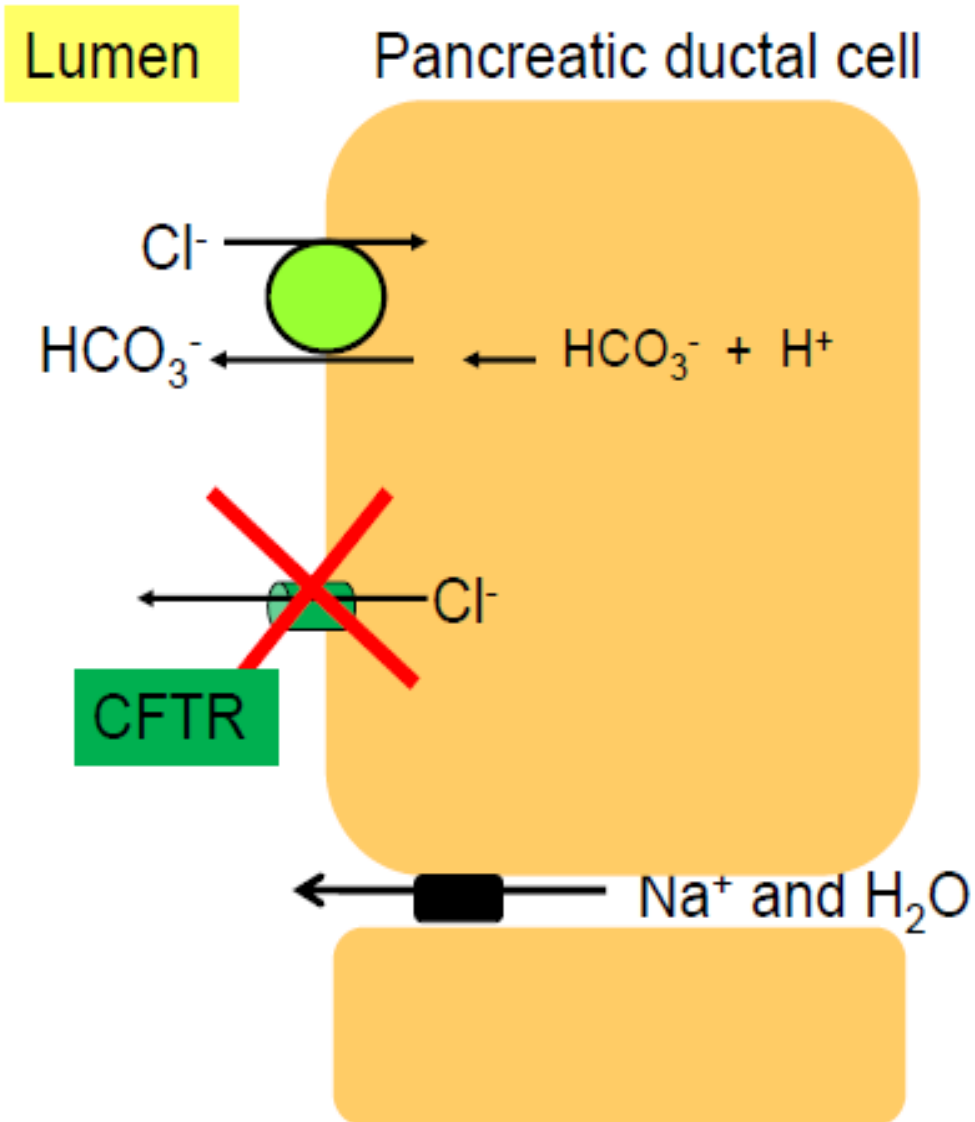


- HCO_3^-
 - produced by carbonic anhydrase
 - secretion via $\text{HCO}_3^-/\text{Cl}^-$ exchanger
- Cl^-
 - cellular Cl^- can diffuse passively back into lumen through CFTR channel
 - NKCC transporter also provides cellular Cl^-
- Na^+ and H_2O
 - follow passively para-cellularly down electrochemical gradient
- H^+
 - formed with HCO_3^- in the cell
 - exits across basolateral membrane via Na-H exchanger, entering bloodstream and neutralizing the alkaline tide

The Pancreatic two component model



Clinical Correlate: Cystic Fibrosis



- Defective CFTR (Cl^- channel and eNaC regulator)
- Consequences:
 - ↓ Cl^- secretion
 - ↓ Lumen negativity
 - ↓ Na^+ and H_2O movement
 - ↓ Ductal secretion
 - ↓ Enzyme action → Malabsorption

Pancreatic secretion time course

PHASE	SECRETION RATE
Interdigestive (Fasting) Phase	Basal
1. Cephalic	↑ (25%)
2. Gastric	↑ (10%)
3. Intestinal	↑ ↑ (50-80%)
Late Digestive	↓

Pancreatic Secretions: Cephalic Phase (25%)

Thought, smell,
taste, chewing,
swallowing of food

CEPHALIC PHASE

**↑ Vagal activity
(ACh)**

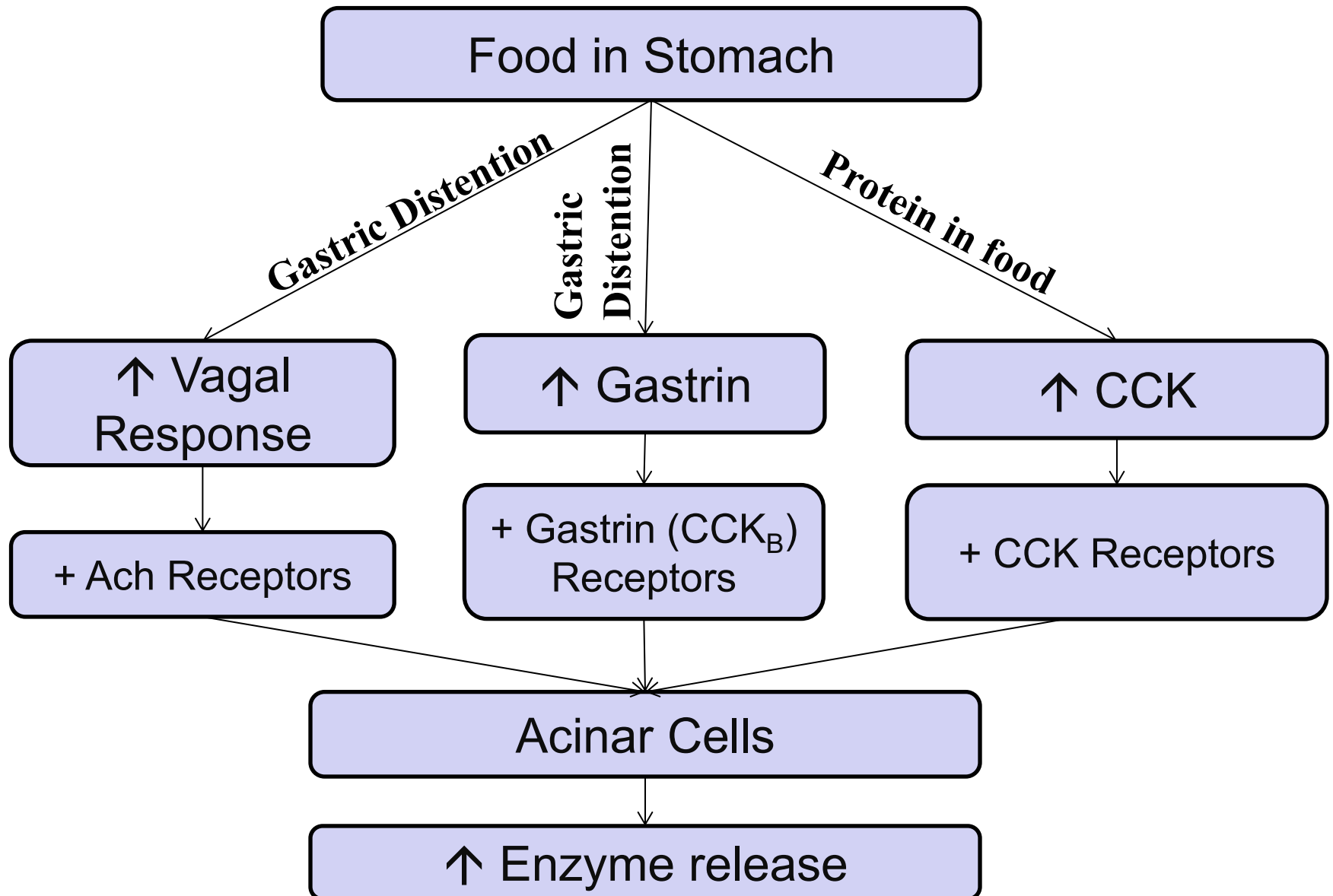
+ Acinar Cells

**↑ Secretion
(*enzymes)**

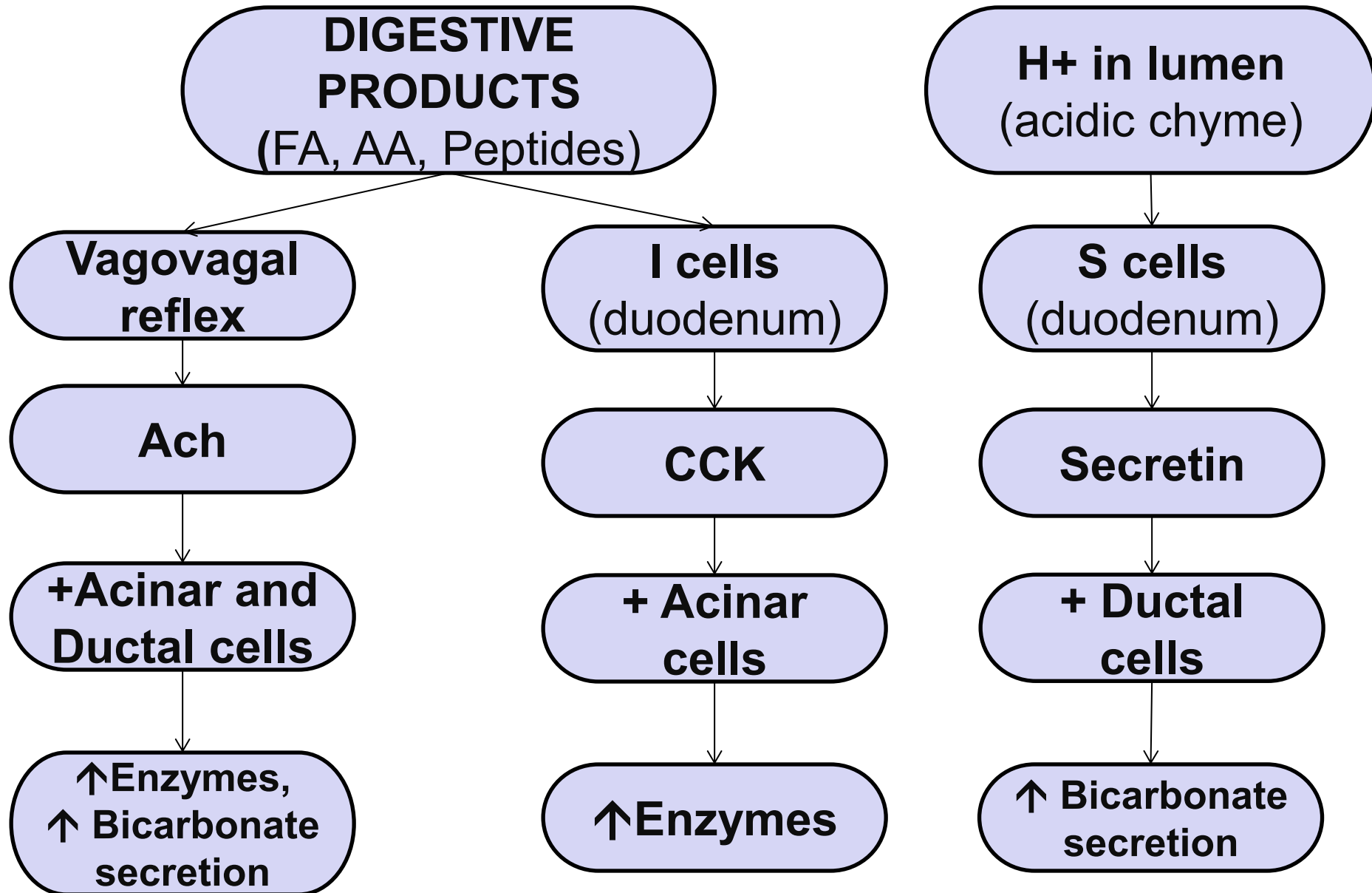
+ Ductal Cells

**↑ Secretion
(HCO₃⁻)**

Pancreatic Secretions: Gastric Phase (10%)



Pancreatic Secretions: Intestinal Phase (50-80%)



Intestinal Phase - CCK

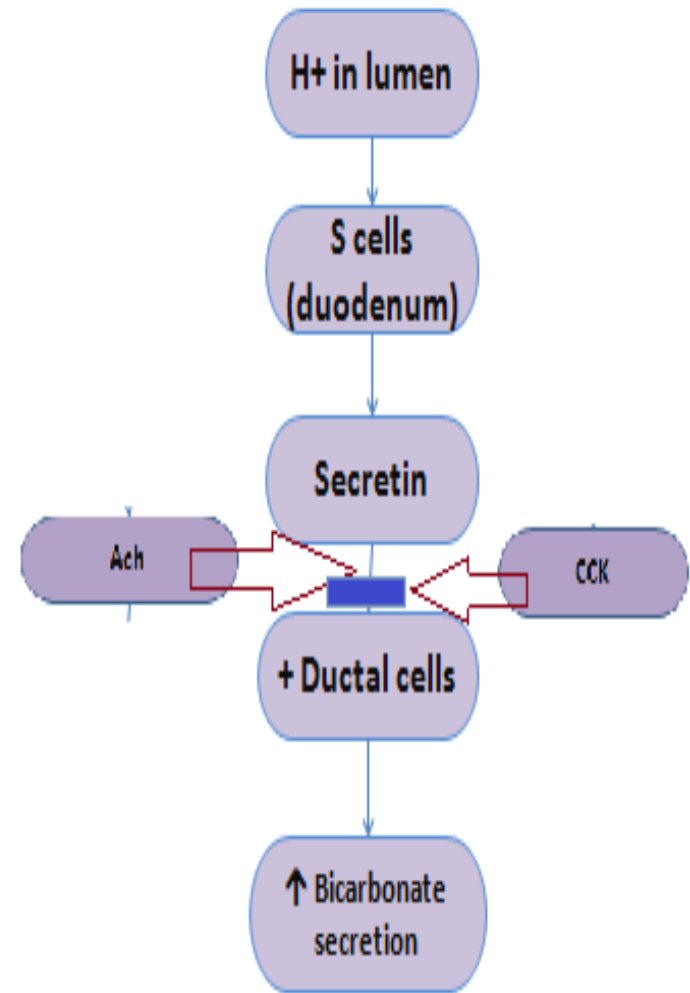
- CCK release depends on content of meal
 - Fatty acids: large CCK response
 - Amino Acids: intermediate CCK response
 - Carbohydrates: minimal CCK response
- Exocrine pancreas has large reserves of digestive enzymes for carbohydrates and protein, less for lipid digestion
- Even lipid digestion is not affected until 80-90% of the pancreas is destroyed

Endocrine Regulators: Cholecystokinin (CCK)

- **Key role:** control of gastric emptying and pancreatic enzyme release
- **Source:** I cells in duodenum and jejunum
- **Stimuli for release:**
 - Small peptides, amino acids, fatty acids, monoglycerides
- **Inhibition of release:**
 - Absence of stimuli (via absorption)
 - Somatostatin
- **Effects:**
 - Stimulates pancreatic enzymes and HCO_3^- secretion
 - Slows gastric emptying to allow more time for intestinal digestion and absorption
 - Causes gallbladder contraction → increases bile duct pressure → sphincter of Oddi relaxes → bile flows into duodenum (pain occurs if gallstones obstruct this process)
 - Relaxes sphincter of Oddi

Secretin

- acts on pancreatic duct cells to increase cAMP levels, activating HCO_3^- and fluid secretion
- food buffers gastric acidity
 - pH drop in duodenum may not be enough to produce high secretin levels
- CCK and ACh potentiate the effect of secretin on ductal HCO_3^- and fluid secretion



Endocrine Regulators: Secretin

- **Key role:** Control of pancreatic HCO_3^- secretion
- **Source:** S cells in duodenum
- **Stimuli for release**
 - H^+ in the lumen of the duodenum
 - Fatty acids and peptides in the duodenal lumen
- **Inhibitors**
 - Neutral pH in duodenum
 - Somatostatin
- **Effects**
 - Stimulates pancreatic HCO_3^- and H_2O secretion -> neutralizes small intestine acidity from stomach acid
 - Inhibits H^+ secretion by gastric parietal cells
 - Stimulates HCO_3^- secretion from bile ducts.

Summary: Pancreatic regulation

Small increases in pancreatic and hepatic secretions may occur during the cephalic and gastric phases of control, but most activity in the small intestine occurs as it digests and absorbs a meal.

Phase	Stimulus	Mediators	Response
Cephalic	Thought of food, smell, taste, chewing, swallowing	Release of Ach and gastrin by vagal stimulation	↑ secretion with greater effect on enzyme output
Gastric	Protein in food	Gastrin	↑ secretion - enzymes
	Gastric distension	Ach release by vagus	↑ secretion - enzymes
Intestinal	Acid in chyme	Secretin	↑ Water and HCO_3^- secretion
	Long chain fatty acids	CCK and vagovagal reflex	↑ secretion - enzymes
	Amino acids and peptides	CCK and vagovagal reflex	↑ secretion - enzymes

Part 2

Intestinal secretions

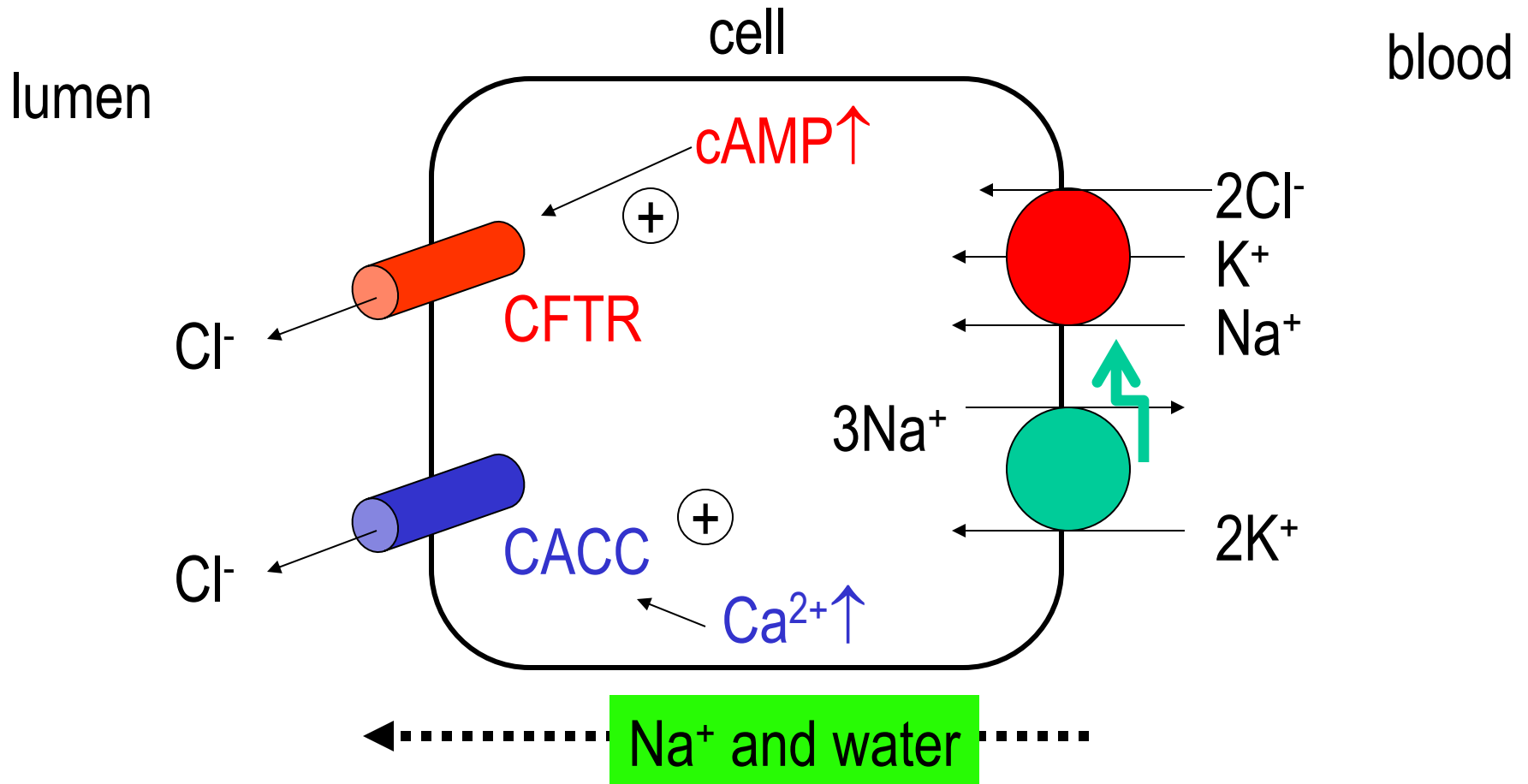
Small Intestine Secretions

- Composition is mainly watery mucous and enzymes
- Functions
 - Protection
 - mucus prevents acid and protease damage to intestine
 - alkaline secretions neutralize chyme and bacterial acids
 - carry gut antibodies from crypts into lumen
 - Motility
 - lubrication enhances motility
 - reduced secretions can cause gut obstructions
 - Digestion/absorption
 - enzymes break down all major foodstuff
 - secretions rich in Na^+ provides a source of Na^+ for absorption
 - Control
 - mainly via local reflexes stimulated by chyme (acidity and distension)

Intestinal fluid and ion balance

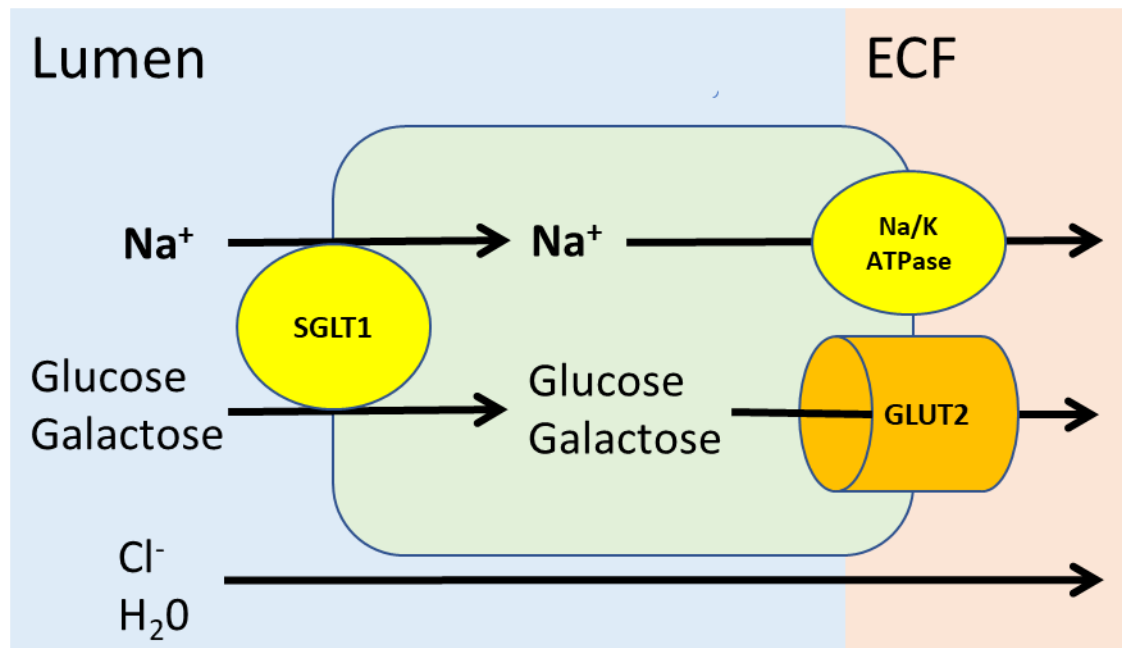
- Please refer back to DLA 4 (Epithelial function) from module FTM for general epithelial function
- Secretions
 - driven by Cl^- release into the lumen
 - secretory mechanisms mainly located in crypts
- Absorption
 - relies on Na^+ uptake from the lumen
 - absorption mechanisms mainly located in the villi
 - summarized in handout for DM 39, please read before the lecture!
- H_2O follows osmotically in both cases

Cl⁻ Transport Drives Fluid Secretion From Crypts



SGLT1 in intestinal absorption

- prominent in small intestine and proximal colon
- apical Na^+ energy gradient drives glucose uptake
- 260 H_2O molecules transported along per cycle
- mediates ~ 5 L/day of water reabsorption

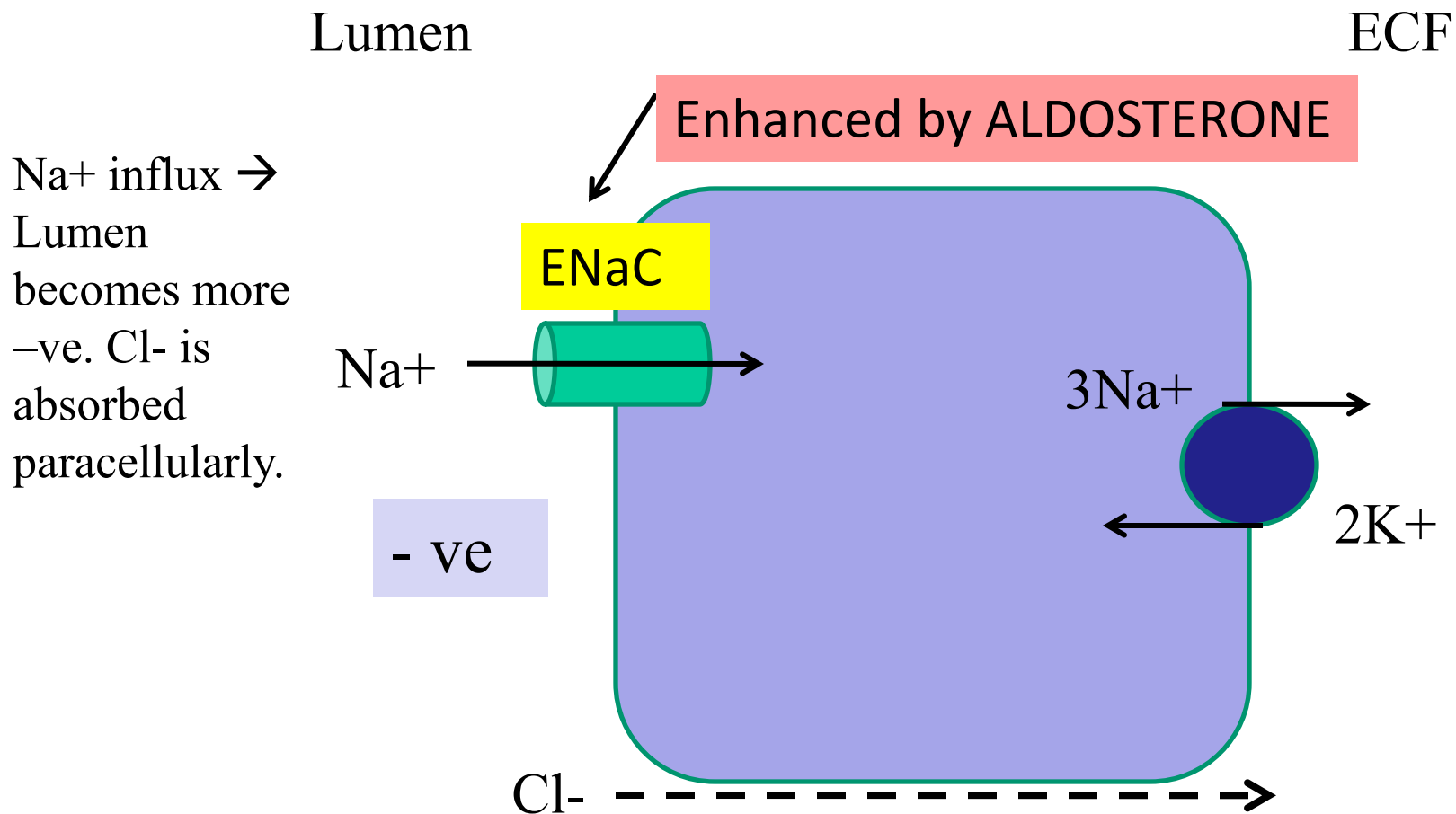


**Oral rehydration
solution contains
both glucose and
sodium-why?**

Large Intestinal secretions

- Composition:
 - Alkaline mucus solution containing bicarbonate and K^+
- Function:
 - Protection
 - Resident bacteria release acid
 - Mechanical damage by passage of feces
- Stimulation:
 - Acid and mechanical
 - Mediated by long and short reflexes

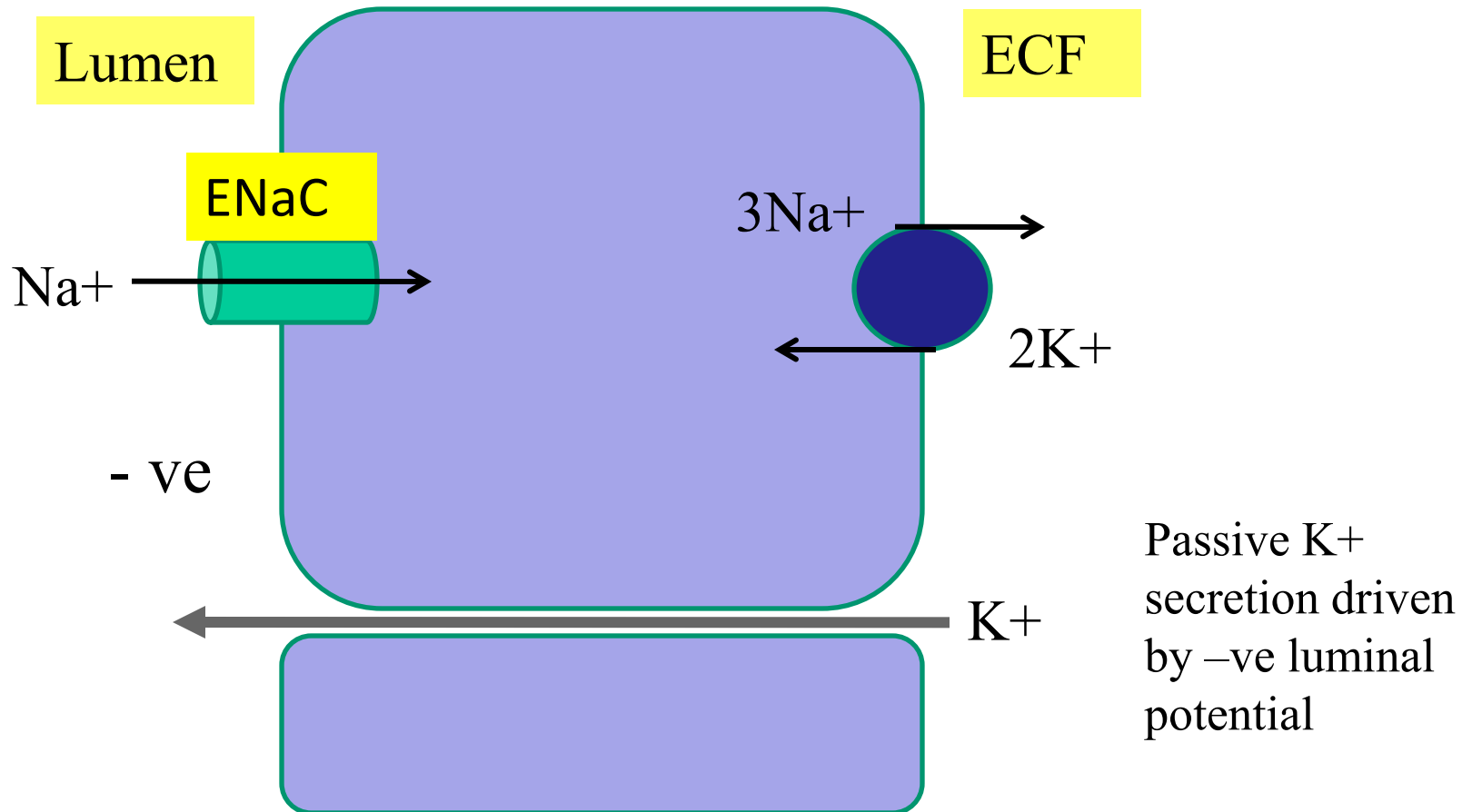
Distal colon NaCl reabsorption



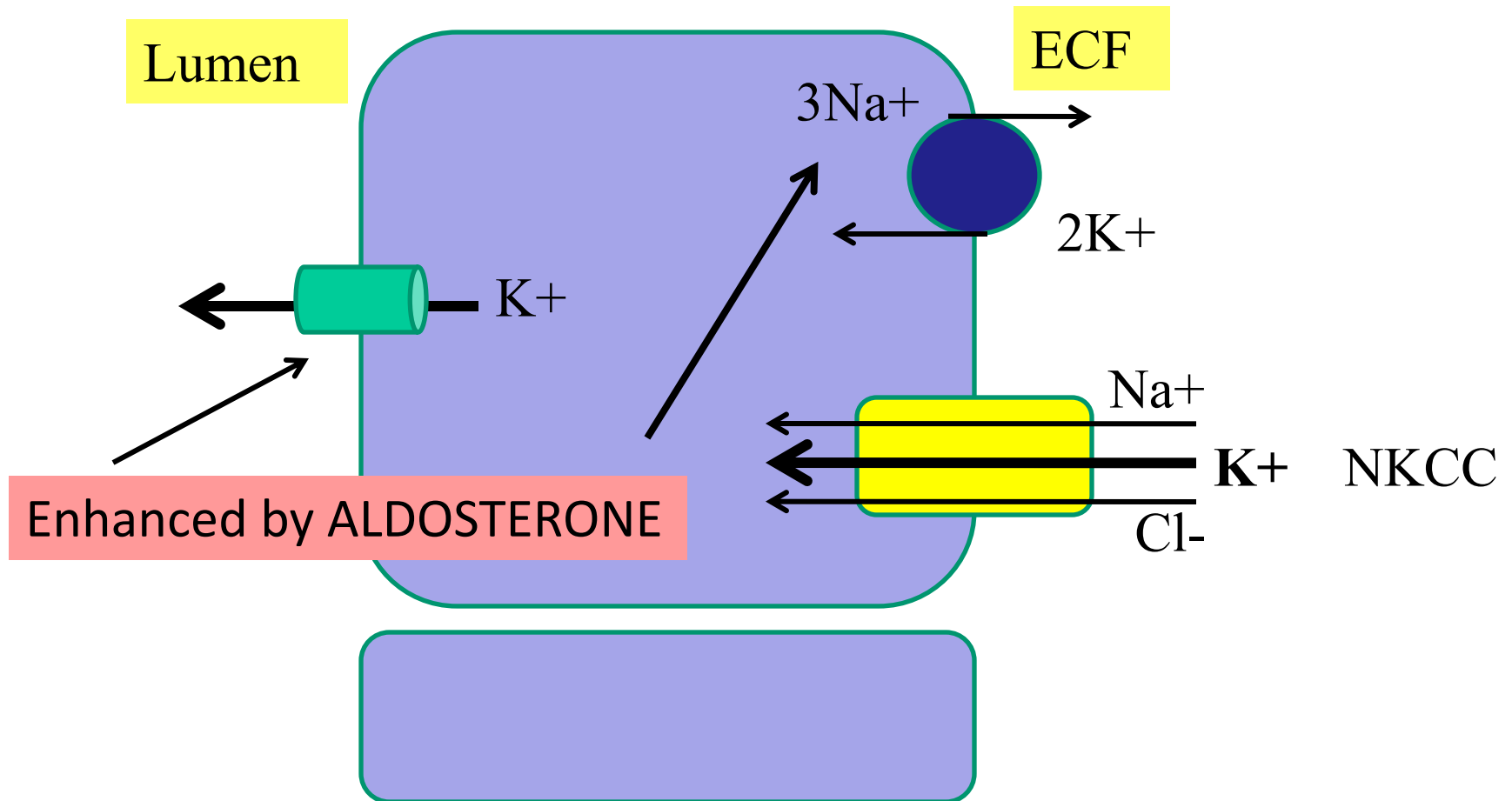
When body Na^+ is low, aldosterone stimulates the ENaC pump to promote Na^+ retention (similar in the kidney).

Secretion of K^+ in the distal colon

1. Passive K^+ secretion



Active K^+ secretion in the distal colon



K⁺ movement summary

- Intestines can absorb or secrete K⁺
 - GIT plays a minor role in regulating K⁺ in the ECF.
 - The kidney is the major regulator of K⁺ levels.
- K⁺ is absorbed in proximal segments of the GIT
- K⁺ is secreted in distal segments of the GIT
- Salivary, Gastric, Pancreatic and Biliary secretions: rich in K⁺
- High K⁺ load is presented to the small intestine
- Stool contains high levels of K⁺ (partly due to colonic secretion)

Intestinal secretion & absorption summary

Small Intestine

- Net absorption
 - H_2O
 - Na^+
 - Cl^-
 - K^+
- Net secretion
 - HCO_3

Large Intestine

- Net absorption
 - H_2O
 - Na^+
 - Cl^-
- Net secretion
 - K^+

Part 3

GI fluid balance and diarrhea

GI fluid balance

SECRECTIONS

Diet: 1 – 2 L
Saliva: 1.5 L

Gastric Juice: 2.0 L

Pancreatic Juice: 1.5 L

Bile 0.5 L

Small Intestine 1 L

Colon

Fluid Load to SI =
7 – 9 L / day

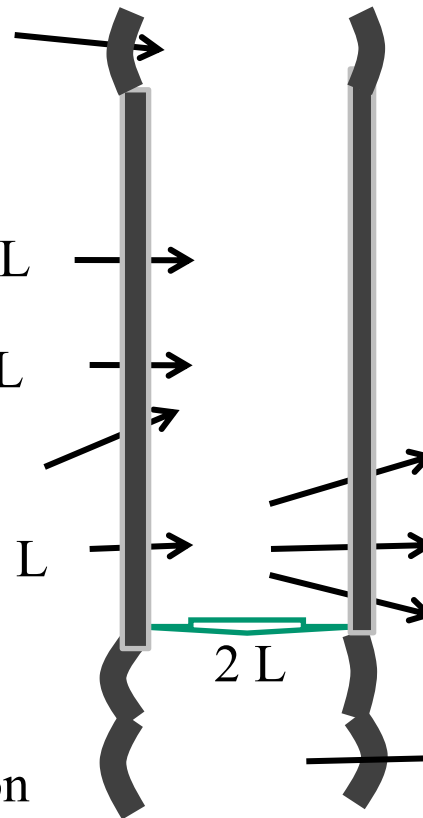
ABSORPTION

6.5 L (SI)

1.9 L Colon

Fluid Absorbed =
98%

Stool water:
0.1 L



Diarrhea

Definition: an increase in the volume or frequency of defecation.

Two types: Osmotic and Secretory diarrhea.

- Osmotic

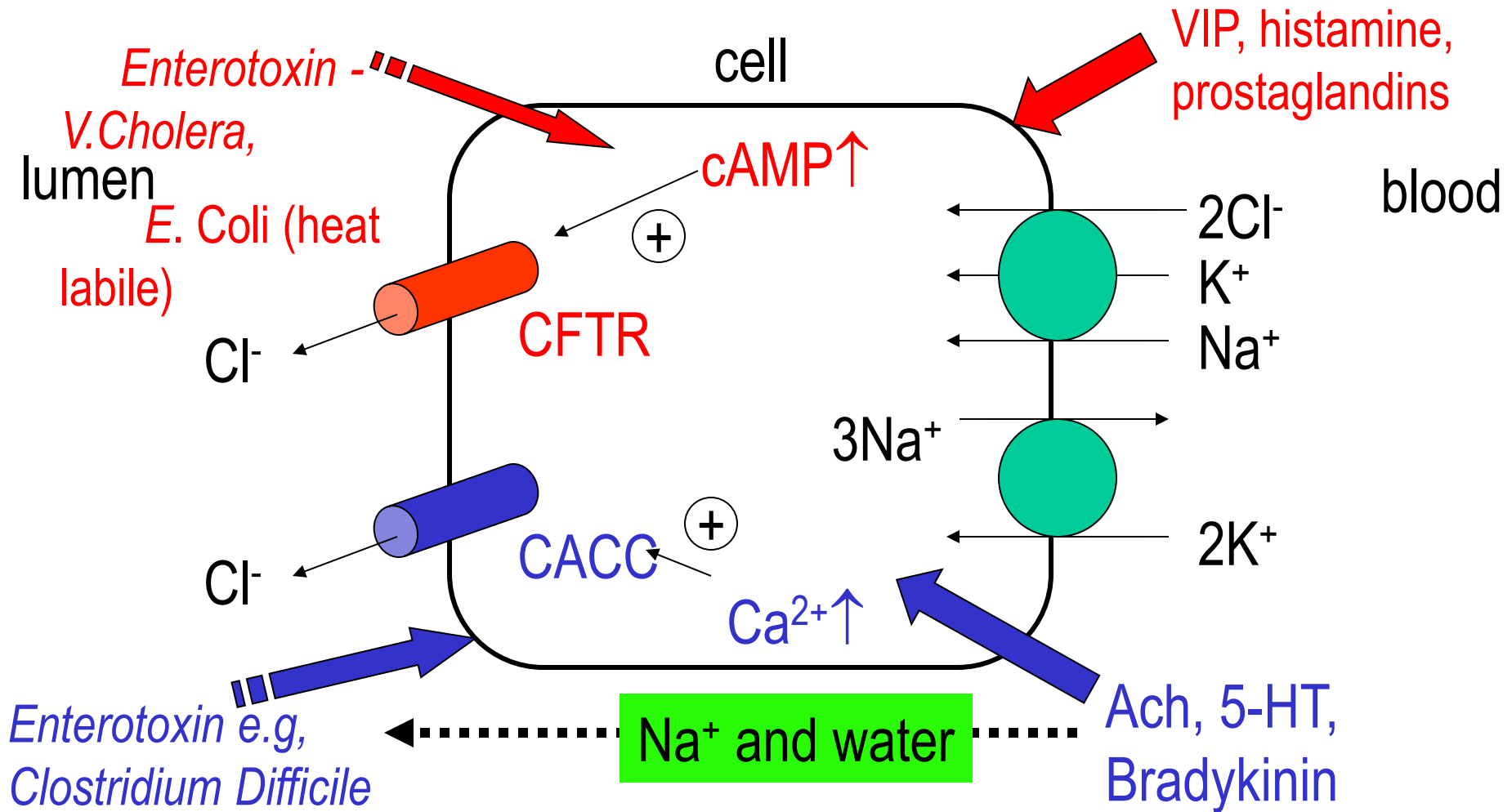
- Presence of poorly absorbed substrates in the intestine.
- Example: Lactose intolerance

- Secretory

- Increased intestinal secretion of fluids and electrolytes.
- Examples: Bacterial toxins, Traveler's diarrhea, Tumors secreting certain hormones

Secretory Diarrhea

(Increased secretions of fluid and electrolytes from GIT)



Factors affecting Cl⁻ secretion in SI

Apical Cl^- channels open in response to:

- ↑ cAMP
- ↑ Ca^{2+}

Secretagogues induce apical Cl channel opening

Secretagogues: Hormones and neurotransmitters (ENS, VIP)

Either by Products of immune cells (e.g. histamine)

↑cAMP

or

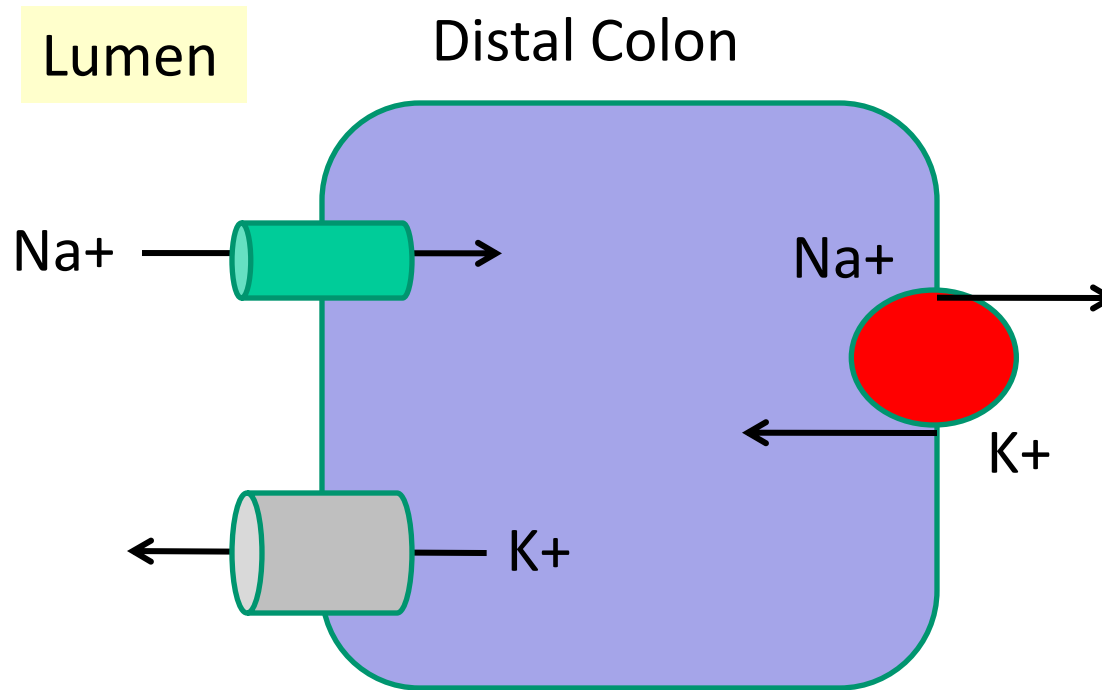
$$\uparrow \text{Ca}^{2+}$$

Bacterial exotoxins (“enterotoxins”)

Laxatives

Induction (recruitment and activation) of apical Cl⁻ channels plays a major role in the pathophysiology of secretory diarrheas.

Diarrhea and potassium balance



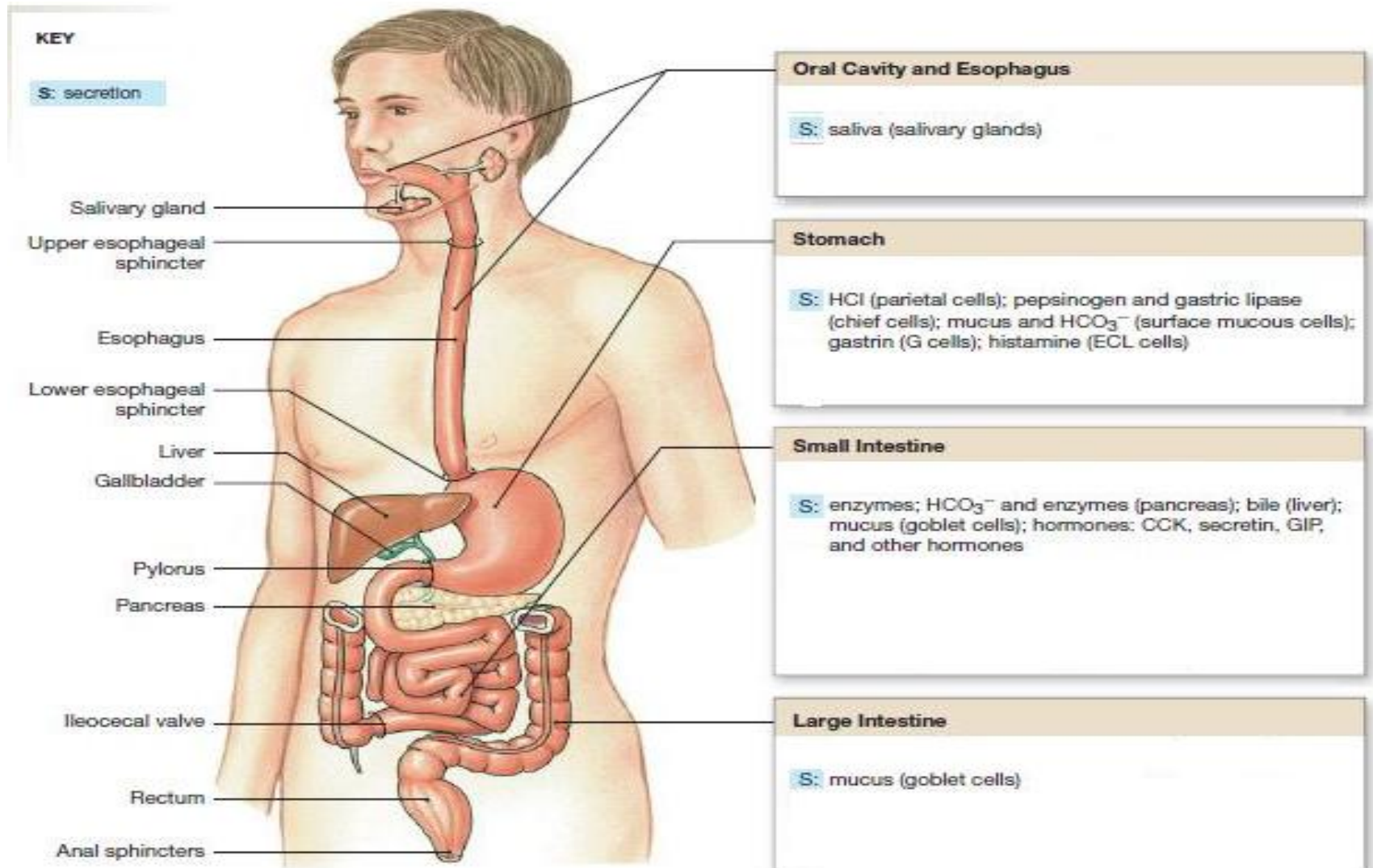
↑ Rate of fluid flow during diarrhoea results in net loss of K⁺

Loss of K⁺ → HYPOKALEMIA ($[K^+] < 3.5 \text{ mmol/L}$)

Summary: At this point, you should be able to

- Map the various secretions of each part of the GIT
- Recall the composition of each secretion
- Understand the function of each secretion
- Predict the changes in secretion with certain stimuli

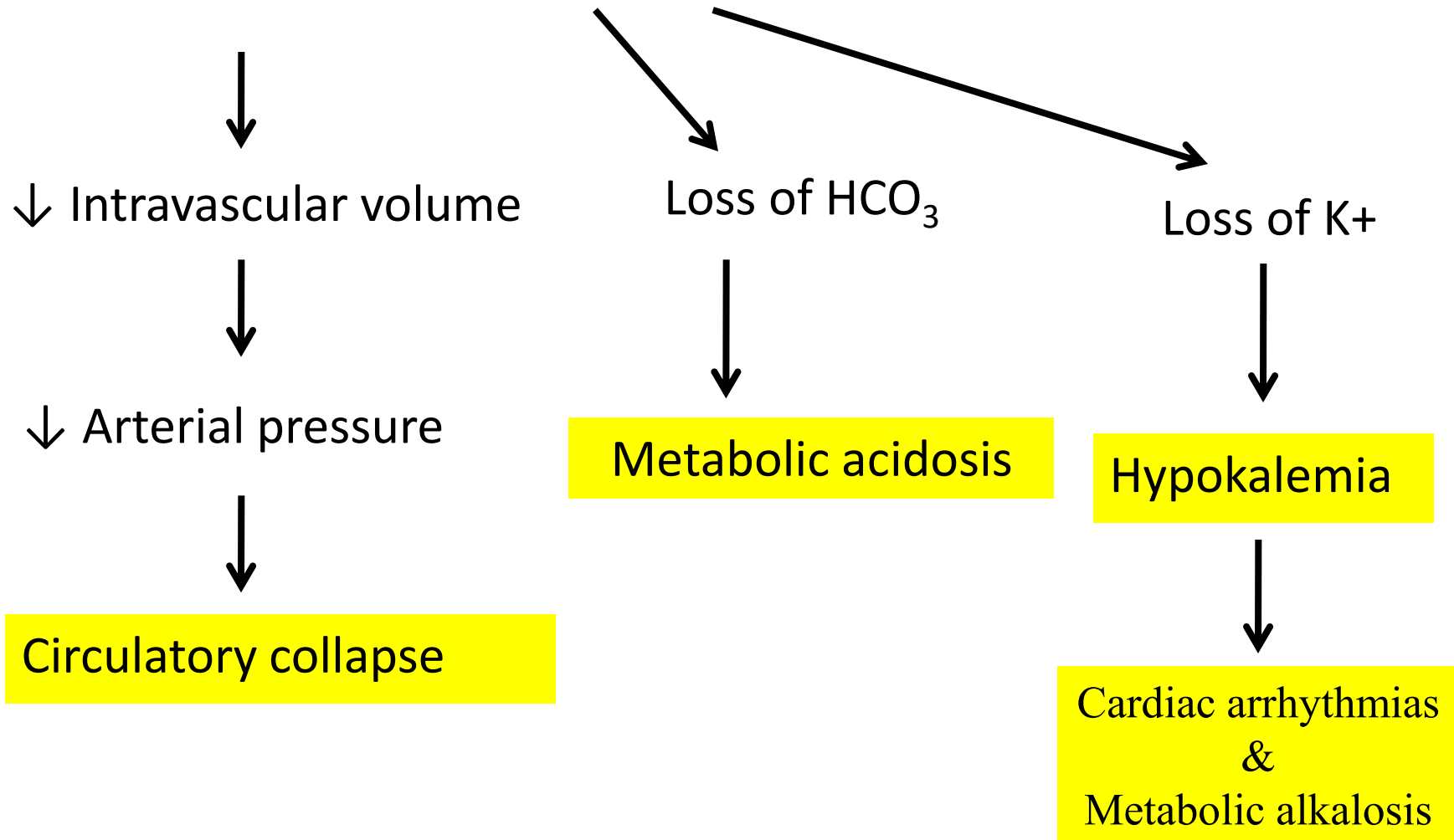
Study slide: Secretions- Add to this diagram the various compositions, functions and stimulatory factors



Consequences of Secretory Diarrhea

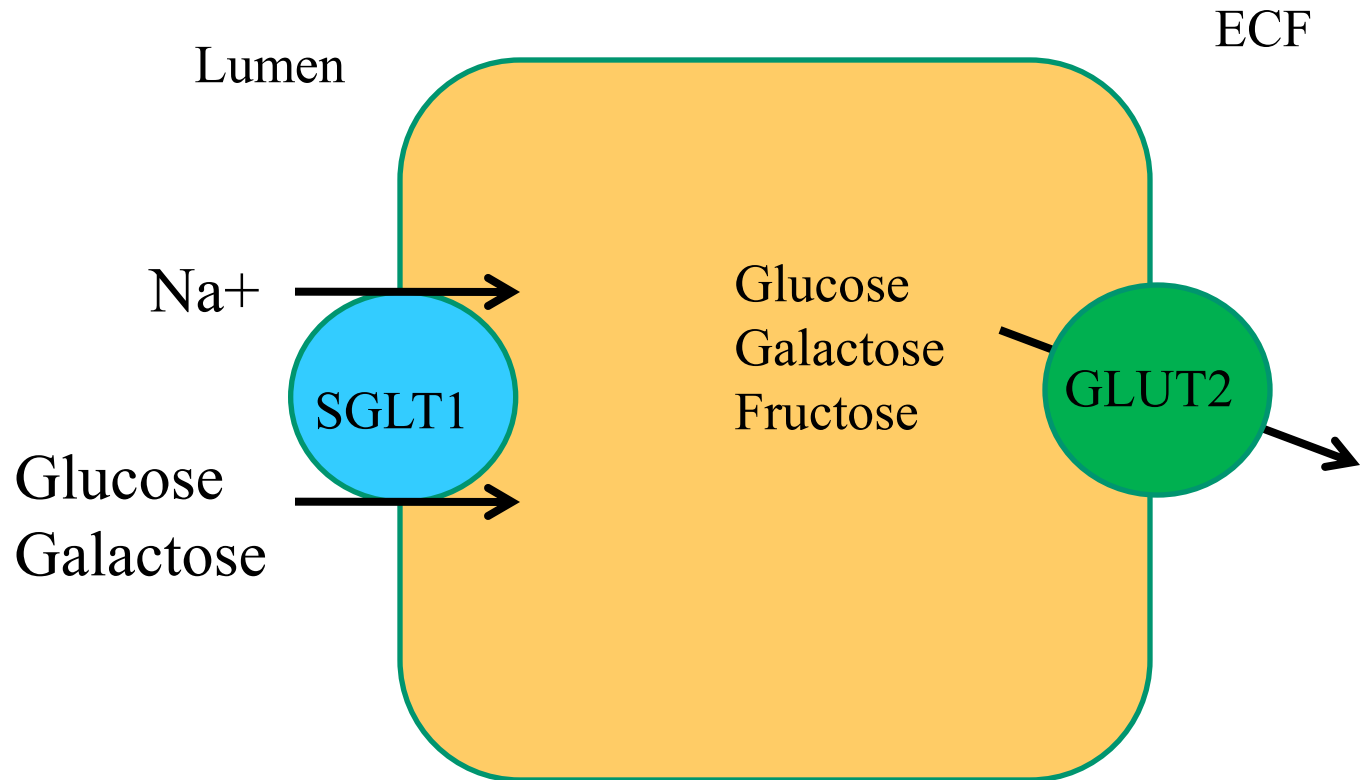
Summary study slide

Loss of ECF-type fluid from the GIT: (up to 9L/day is possible!)



Oral Rehydration Therapy: Solution with Na^+ and Glucose

H_2O follows by osmosis



Absence of SGLT 1 transporter:

Glc and Galactose malabsorption

→ Osmotic retention of water in GIT

→ Diarrhea

Practice question slide

Which of the following statements are correct?

-  1. H^+ ions are pumped into the stomach against a concentration gradient
- 2. Histamine increases cAMP and reduces acid production
-  3. The vagus stimulates acid production directly and via gastrin release
-  4. Prostaglandins inhibit acid secretion
- 5. Gastrin production is enhanced when the stomach contents pH are < 3

Important gastrointestinal hormones

Hormone	Actions	Secretion site
Gastrin	•↑ Gastric H^+ secretion	G cells (gastric antrum, duodenum)
Somatostatin	•↓ Secretion of most GI hormones	D cells (pancreatic islets, gut mucosa)
Cholecystokinin	•↑ Pancreatic enzyme & HCO_3^- secretion	I cells (small intestine)
Secretin	•↑ Pancreatic HCO_3^- secretion •↓ Gastric H^+ secretion	S cells (small intestine)
GIP	•↑ Insulin release •↓ Gastric H^+ secretion	K cells (small intestine)
Motilin	•↑ GI motility	M cells (small intestine)
GI = gastrointestinal; GIP = gastric inhibitory peptide; H^+ = hydrochloric acid; HCO_3^- = bicarbonate.		